

Thomas LOUIS-SARROLA

PhD Student in Statistical Physics

thomas.louissarrola@gmail.com

+33 7 86 53 76 37

lsthomas.github.io

EDUCATION

PhD in Statistical Physics

2025–2028

Université Paris-Saclay, Orsay

High-dimensional dynamics: statistics of equilibria and dynamical trajectories. Supervised by Valentina Ros (CNRS) and Pierfrancesco Urbani (CNRS).

M2 in Complex Systems Physics

2024–2025

Sorbonne Université, Paris

Graduated with highest honours; awarded a doctoral-school scholarship in physics for the Île-de-France region.

M2 in Probability and Finance

2023–2024

Sorbonne Université, Paris

École Polytechnique Engineering Programme

2020–2024

École Polytechnique, Paris

M1 in Fundamental Mathematics.

EXPERIENCE

Quantitative Researcher

2025

Qube Research & Technologies, Paris

Six-month internship supervised by Augustin Landier (HEC Paris).

Research Intern in Mathematical Physics

2023

DMA, École Normale Supérieure, Paris

Four-month internship supervised by Isabelle Gallagher on deterministic and stochastic Navier–Stokes equations. Recipient of the 2023 research internship prize.

Research and Development Intern

2022

ID Quantique, Geneva

Three-month internship researching classical and quantum random number generators.

TEACHING

Advanced Mathematics

2026–2027

Sciences Po, Paris

Tutorials, including the pre-term intensive course.

Algorithms for Discrete Mathematics (APM_2F005_EP)

2026–2027

École Polytechnique, Palaiseau

Practical sessions.

Aléatoire (APM_3X061_EP)

2026–2027

École Polytechnique, Palaiseau

Tutoring.

Analysis, Bachelor 1A

2025–2026

École Polytechnique, Palaiseau

Tutorials (64 hours), taught with Frank Pacard.

PUBLICATIONS

Fragile vs robust multiple equilibria phases in generalized Lotka–Volterra model with non-reciprocal interactions.

Thomas Louis-Sarrola and Valentina Ros. *SciPost Physics* **21**, 014 (2026). doi:10.21468/SciPostPhys.21.1.014.

CONFERENCES AND SCHOOLS

Fundamental Problems in Statistical Physics XVI

2025

Oropa, Italy

International summer school.

2nd Barcelona School of Non-Equilibrium Physics

2026

Barcelona, Spain

Presentation.

Workshop “Random Matrix Theory for Ecology, Economics, Finance and Statistical Physics”

2025

ZiF, Bielefeld, Germany

Journées françaises des Systèmes Complexes 2026

2026

Lyon, France

Presentation.

INTERESTS AND SKILLS

Sport

Basketball (captain of École Polytechnique’s team); road and trail running.

Programming

Python, OCaml, Java, LaTeX.

Languages

French (native), English (fluent; international baccalaureate), German (intermediate), Corsican (basic).